

Acquisition Assessment: Insmmed Incorporated

Acquisition Assessment for the Strategy Committee

21 July 2026

Acquisition Assessment: Insmmed Incorporated (NASDAQ: INSM)

1. Executive Summary

- **Target Profile:** Insmmed Incorporated (NASDAQ: INSM) is a commercial-stage biopharmaceutical leader focused on rare respiratory and neutrophilic inflammatory diseases, headquartered in Bridgewater, New Jersey.
 - **Anchor Assets:** Portfolio is anchored by **BRINSUPRI™ (brensocatib)**, a first-in-class oral dipeptidyl peptidase 1 (DPP1) inhibitor FDA-approved in August 2025 for non-cystic fibrosis bronchiectasis (NCFB); **ARIKAYCE®** (amikacin liposome inhalation suspension), an established therapy for refractory *Mycobacterium avium* complex (MAC) lung disease transitioning into frontline treatment following positive Phase 3b ENCORE data; and **Treprostinil Palmitil Inhalation Powder (TPIP / INS1009)**, a once-daily inhaled lipophilic prodrug in Phase 3 development for Pulmonary Arterial Hypertension (PAH) and PH-ILD.
 - **Deal Thesis Archetype: Commercial Tuck-In & Late-Stage Category Dominance Platform.** Acquiring Insmmed immediately establishes an unassailable commercial monopoly in NCFB and NTM lung infections while securing a potential multi-billion dollar class-disruptive once-daily inhaled prostacyclin for pulmonary vascular disease.
 - **Single Biggest Value Driver:** Commercial scale-up of BRINSUPRI in NCFB (estimated \$3.96B U.S. peak sales / \$5.36B global peak) combined with TPIP's Phase 3 disruption of the \$2.0B+ inhaled prostacyclin market by replacing 4x-daily Tyvaso with a once-daily dry powder inhaler.
 - **Single Biggest Risk:** Heavy near-term commercial launch burn and ongoing Phase 3 clinical costs coupled with a tight cash runway window of **16.5 months** (\$1.22B combined liquidity vs \$222.74M Q1 2026 quarterly operating cash burn), requiring balance sheet integration, alongside potential Freedom-to-Operate (FTO) litigation from United Therapeutics regarding TPIP dry-powder inhaler delivery mechanics.
 - **Recommendation: PURSUE WITH CONDITIONS**
 - **Conviction Level: HIGH**
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2. Company & Deal Thesis

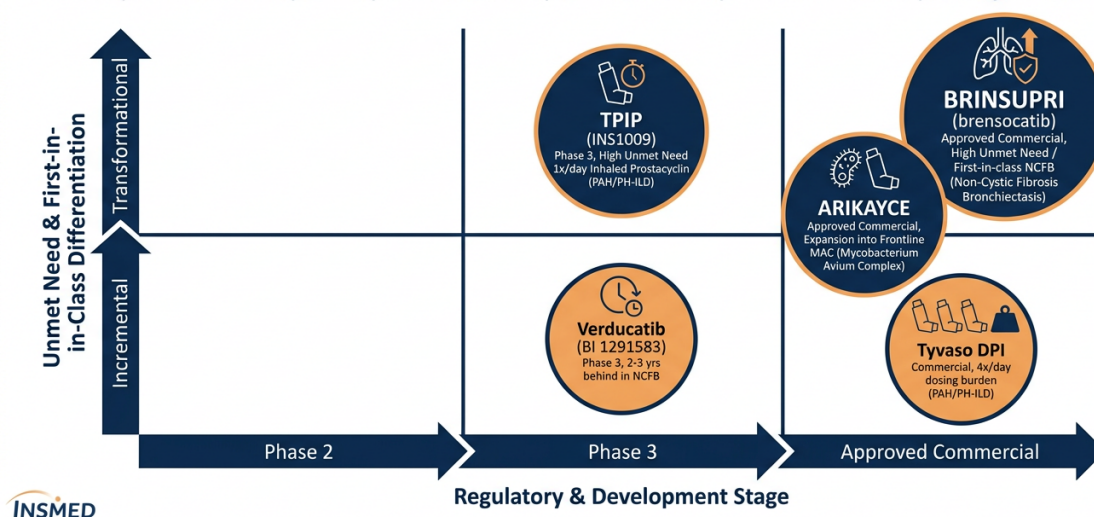
Corporate Context & Strategic Stage

Insmmed Incorporated has transitioned from a single-product rare disease company into a fully integrated, multi-asset commercial biopharmaceutical leader. The company's commercial engine generated \$606.42M in FY2025 net revenue (\$433.80M ARIKAYCE + \$172.62M initial BRINSUPRI launch contribution), expanding to \$305.96M in Q1 FY2026 (Form 10-Q, Q1 FY2026).

Despite commercial success, Insmed faces a pivotal strategic crossroads. Full-scale global commercial launch of BRINSUPRI across North America, Europe, and Japan, concurrent with global Phase 3 pivotal programs for TPIP (PALM-PAH and PALM-ILD), consumed \$771.09M in R&D and \$935.01M in operating cash burn in FY2025 (Form 10-K, FY2025). With combined liquidity of \$1.22B as of March 31, 2026, Insmed faces a financing window wall within 16.5 months. This capital intensity creates a prime acquisition window for a large-cap acquirer to integrate Insmed's high-margin assets onto an established global commercial infrastructure.

Insmed dominates NCFB and NTM while disrupting PAH/PH-ILD.

A competitive landscape analysis of Insmed's portfolio vs competitors in rare respiratory diseases.



Deal Thesis & Strategic Synergy

- Uncontested NCFB First-Mover Advantage:** BRINSUPRI is the first and only FDA-approved disease-modifying therapy for NCFB, an addressable U.S. patient population of ~500,000. Key pipeline competitors remain 2 to 3 years behind in Phase 3.
- NTM Market Expansion:** ARIKAYCE's successful Phase 3b ENCORE study de-risks frontline MAC expansion, expanding addressable U.S. patient prevalence by 3.5x to 5x (from ~20,000 refractory patients to >100,000 total MAC patients).
- Best-in-Class PAH Class Disruption:** TPiP delivers true 24-hour hemodynamic coverage via a once-daily capsule dry powder inhaler (DPI), directly targeting United Therapeutics' market-leading Tyvaso/Tyvaso DPI (\$1.88B FY2025 revenue), which is burdened by a 4x-daily dosing schedule and high cough-related discontinuations.

3. Scientific & Modality Assessment

Lead Asset Biological Rationales

1. Brensocatib (BRINSUPRI™ – DPP1 Inhibitor)

- Mechanism of Action:** Oral, small-molecule, reversible inhibitor of Dipeptidyl Peptidase 1 (DPP1, also known as Cathepsin C).

- **Target Validation:** DPP1 is the essential enzyme in bone marrow progenitor cells responsible for cleavage and activation of neutrophil serine proteases (NSPs: neutrophil elastase [NE], proteinase 3 [PR3], and cathepsin G [CatG]) during myelopoiesis. By inhibiting DPP1 in the bone marrow prior to mature neutrophil release, brensocatib “defangs” systemic neutrophils, halting destructive mucosal lung tissue destruction in NCFB without causing severe immunocompromise or systemic neutropenia (PMC11261371, PMC10189992).
- **Indication Selectivity & Disproof in Cutaneous Disease:** While DPP1 inhibition highly successfully de-risked mucosal respiratory neutrophilia (ASPEN Phase 3), brensocatib failed its Phase 2b CEDAR trial in Hidradenitis Suppurativa (HS) in April 2026 (mean lesion reduction of 40.3%–45.5% vs 57.1% placebo). This confirms that DPP1 biology is strictly effective in mucosal airway neutrophilic inflammation rather than deep systemic cutaneous abscesses and epithelial tunnels.

2. ARIKAYCE® (Amikacin Liposome Inhalation Suspension)

- **Mechanism & Technology:** Inhaled liposomal amikacin delivered via PULMOVANCE™ lipid vehicle technology through the proprietary Lamira® Nebulizer System.
- **Pharmacokinetic Differentiation:** Liposomes are avidly phagocytosed by alveolar macrophages—the intracellular sanctuary where *Mycobacterium avium* complex (MAC) replicates. This targeted local pulmonary delivery achieves drug concentrations in lung tissue up to 50-fold higher than systemic IV aminoglycosides while maintaining low plasma concentrations, eliminating systemic ototoxicity and nephrotoxicity (tri_67437ee74f14).

3. Treprostinil Palmitil Inhalation Powder (TPIP / INS1009)

- **Mechanism & Formulation:** Wholly owned, once-daily inhaled lipophilic prodrug of treprostinil (prostacyclin IP receptor agonist) linked via an ester bond to a 16-carbon alkyl chain.
- **Pharmacokinetic Differentiation:** Unlike native treprostinil, which exhibits rapid pulmonary clearance requiring 4x-daily dosing (Tyvaso), TPIP anchors in lung tissue lipid membranes and undergoes slow, continuous cleavage by local esterases over 24 hours. This delivers sustained pulmonary vasodilation and anti-remodeling activity without the sharp peak-plasma concentration spikes that drive systemic prostacyclin toxicities (flushing, jaw pain, and severe cough).

Competitive Class Crowding Matrix

Asset / Brand	Target Mechanism	Modality / Route	Development Stage	Competitive Position & Class Status	Source / Reference
BRINSUPRI (brensocatib)	DPP1 / Cathepsin C Inhibitor	Oral Tablet (10mg, 25mg)	FDA Approved (Aug 2025); EC Authorized	First-in-class / First-to-market in NCFB. 2–3 year commercial lead.	SEC Form 10-K FY2025; tri_eebeff34b8ae
Verducatib (BI 1291583)	Cathepsin C (DPP1) Inhibitor	Oral Tablet	Phase 3 (AIRTIVITY)	Principal threat. Phase 2 AIRLEAF showed 32% exacerbation drop.	ClinicalTrials.gov NCT05238675

Florensocaticib (<i>HSK31858</i>)	DPP1 Inhibitor	Oral Tablet	Phase 3 (<i>SAVE-BE</i>)	Chiesi ex-China partner; Phase 2 reduced exacerbations in Chinese NCFB.	Lancet Respir Med 2024
ARIKAYCE (<i>LAI</i>)	Aminoglycoside Antibiotic	Inhaled Liposomal Suspension	Approved (Refractory); sNDA H2 2026 (Frontline)	Monopolistic position. Main pipeline competitors (SPR720, epetraborole) failed.	SEC Form 10-K FY2025; tri_af122a9dbc72
SPR720 (<i>Spero Tx</i>)	GyrB Inhibitor	Oral Prodrug	DISCONTINUED	Failed Phase 2a primary endpoint; Grade 3 hepatotoxicity observed.	Spero Update Oct 2024
Epetraborole (<i>AN2 Tx</i>)	Leucyl-tRNA Synthetase	Oral Small Molecule	TERMINATED	Terminated Phase 2/3 trial due to lack of culture conversion efficacy.	AN2 Press Release Aug 2024
TIPIP (<i>INS1009</i>)	Treprostinil Prodrug (IP Agonist)	Inhaled Dry Powder (1x/day)	Phase 3 (<i>PALM-PAH / PALM-ILD</i>)	Potential best-in-class. Solves 4x/day dosing of Tyvaso with +55.7m 6MWD gain.	SEC Form 10-K FY2025; tri_a065dea80387
Tyvaso / Tyvaso DPI	Treprostinil IP Agonist	Inhaled Solution / DPI (4x/day)	Approved (\$1.88B FY2025 Rev)	Market leader in PAH/PH-ILD; burdened by 4x daily dosing schedule.	United Therapeutics FY2025
Seralutinib (<i>Gossamer</i>)	PDGFR/c-KIT Kinase Inhibitor	Inhaled Dry Powder	Phase 3 FAILED	Missed primary 6MWD endpoint in Phase 3 PROSERA (Feb 2026).	Gossamer Release Feb 2026

4. Clinical & Regulatory Assessment

Master Pipeline Assessment Matrix

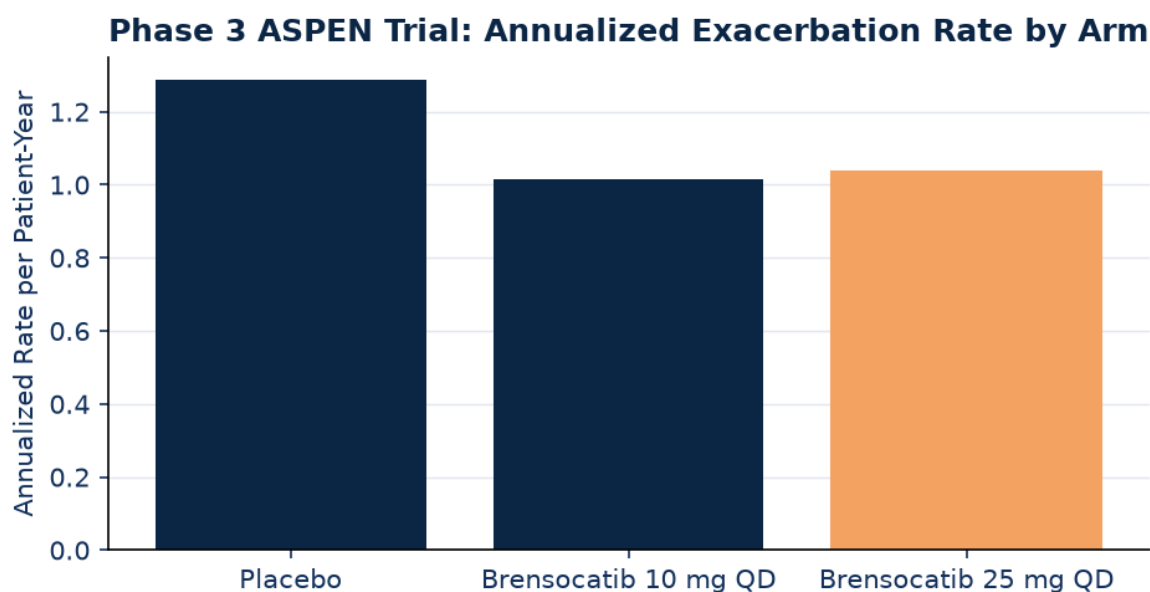
Asset	Target / Indication	Clinical Phase	Key Pivotal Trial Data & Endpoint Status	Regulatory Designations	Next Catalyst & Milestone
Brensocatib (BRINSUPRI)	Non-Cystic Fibrosis Bronchiectasis (NCFB)	Approved (U.S., EU, UK)	Met Primary Endpoint: Phase 3 ASPEN (N=1,721). 10 mg reduced exacerbation rate by 21.1% (p=0.0019); 25 mg by 19.4% (p=0.0046). Slowed FEV1 decline (+38 mL, p=0.04).	FDA Breakthrough, Priority Review, Orphan Drug (US/EU)	Commercial launch expansion in EU/Japan; Japan regulatory decision H2 2026.
Brensocatib	Hidradenitis Suppurativa (HS)	Phase 2b Discontinued	Missed Primary Endpoint: Phase 2b CEDAR (N=214). Active AN reduction (40.3%–45.5%) failed vs placebo (57.1%).	Fast Track	Program discontinued April 2026.
ARIKAYCE	Refractory MAC Lung Disease	Approved (U.S., EU, JP)	Met Primary Endpoint: Phase 3 INS-212 (N=336). 29.0% culture conversion vs 8.9% GBT control (p<0.001).	Accelerated Approval (Subpart H), Breakthrough, Fast Track, Orphan	Traditional approval conversion; post-marketing requirements fulfilled.

ARIKAYCE	Frontline / Treatment-Naïve MAC	Phase 3b Complete	Met Primary Endpoint: Phase 3b ENCORE (N=425). +3.11 pt symptom score diff (p=0.0299); Month 6 conversion 87.8% vs 57.0% (p<0.0001); Month 15 durable 76.2% vs 47.6% (p<0.0001).	Fast Track	sNDA submission to U.S. FDA and PMDA filing in Japan in H2 2026.
TPIP (INS1009)	Pulmonary Arterial Hypertension (PAH)	Phase 3 (PALM-PAH)	Met Primary Endpoint: Phase 2b (N=102). 35% placebo-adjusted PVR reduction (p<0.001); +35.5m 6MWD (p=0.003). 12-Mo OLE: +55.7m 6MWD gain, -60% NT-proBNP.	Fast Track, Orphan Drug	Phase 3 PALM-PAH trial enrollment completion and readout (2027).
TPIP (INS1009)	PH associated with ILD (PH-ILD)	Phase 3 (PALM-ILD)	Met Primary Endpoint: Phase 2a (N=39). Safety/tolerability demonstrated; 10.3% clinical worsening on TPIP vs 50.0% on placebo.	Fast Track	Phase 3 PALM-ILD pivotal trial initiation.
INS1201 / INS1202	DMD / ALS Gene Therapy	Phase 1	Preclinical 10x–50x dose reduction via intrathecal AAV9 delivery. Phase 1 safety dosing initiated (2025–2026).	Orphan Drug	Phase 1 ASCEND (DMD) and ARMOR (ALS) early safety cohorts.

Detailed Trial Design & Efficacy Analysis

Brensocatib ASPEN Phase 3 Trial Analysis

The landmark Phase 3 ASPEN trial (NCT04594369) evaluated oral brensocatib (10 mg QD and 25 mg QD vs placebo) over 52 weeks in 1,721 patients (1,680 adults, 41 adolescents) across 391 global sites (PMC11261371, tri_40139b60eb20). - **Primary Endpoint:** Both dose strengths achieved statistically significant reductions in annualized pulmonary exacerbations (PE): - Placebo: 1.286 PEs/year - Brensocatib 10 mg QD: 1.015 PEs/year (21.1% reduction, Rate Ratio = 0.79, $p = 0.0019$) - Brensocatib 25 mg QD: 1.036 PEs/year (19.4% reduction, Rate Ratio = 0.81, $p = 0.0046$) - **Secondary Endpoints:** - Time to first PE delayed: 10 mg HR = 0.81 ($p = 0.02$); 25 mg HR = 0.83 ($p = 0.04$). - Exacerbation-free at 52 weeks: 48.5% in both active arms vs 40.3% on placebo ($p = 0.02$ and $p = 0.04$). - Lung function preservation: Brensocatib 25 mg slowed post-bronchodilator FEV1 decline over 52 weeks (24 mL loss vs 62 mL loss for placebo; treatment difference +38 mL, $p = 0.04$).



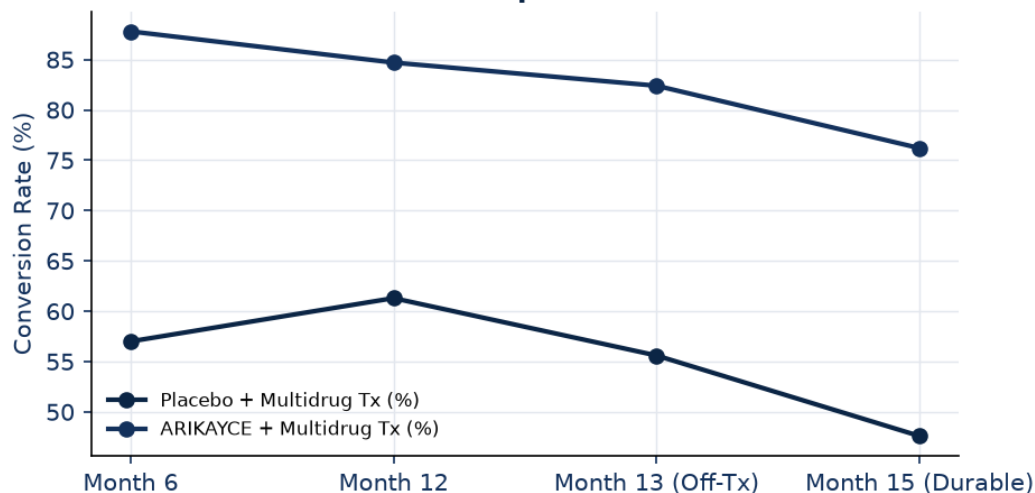
- **Safety & Target Safety De-Risking:**

- Target-related dermatologic adverse events (hyperkeratosis/desquamation) occurred in 1.4% (10 mg) and 3.0% (25 mg) vs 0.7% (placebo), predominantly mild and reversible.
- Periodontitis/gingival events showed no evidence of progressive periodontal breakdown (1.4% 10 mg, 2.1% 25 mg vs 2.7% placebo).
- Pneumonia rates were lower in brensocatib arms (4.0% 10 mg, 4.7% 25 mg) than placebo (5.9%).
- Discontinuation rates due to AEs were low and balanced (4.3% 10 mg, 3.8% 25 mg vs 4.1% placebo).

ARIKAYCE ENCORE Phase 3b Frontline Trial Analysis

The global Phase 3b ENCORE study (NCT04677543) evaluated ARIKAYCE (590 mg once daily) + background multidrug therapy vs placebo + multidrug therapy in 425 newly diagnosed, treatment-naïve MAC patients over 12 months with 3 months off-treatment follow-up (tri_af122a9dbc72). - **Primary Symptom Endpoint:** Statistically significant improvement in Respiratory Symptom Score (QoL-B RSS) at Month 13 (+17.77 ARIKAYCE vs +14.66 placebo; treatment difference +3.11 points, $p = 0.0299$). - **Multiplicity-Controlled Culture Conversion (Secondary Endpoints):** - Month 6 Conversion: 87.8% ARIKAYCE vs 57.0% Placebo ($p < 0.0001$). - Month 12 Conversion: 84.7% ARIKAYCE vs 61.3% Placebo ($p < 0.0001$). - Month 13 Conversion (1 Mo off-treatment): 82.4% ARIKAYCE vs 55.6% Placebo ($p < 0.0001$). - Durable Conversion at Month 15 (3 Mo off-treatment): 76.2% ARIKAYCE vs 47.6% Placebo ($p < 0.0001$).

ARIKAYCE Phase 3b ENCORE Trial: Sputum Culture Conversion Rates (%)



- **Regulatory Conversion Impact:** ENCORE fulfills FDA post-marketing requirements under Subpart H accelerated approval. sNDA submission in H2 2026 will support label expansion to frontline MAC, increasing addressable patient prevalence by 3.5x to 5x.

TPIP (INS1009) Phase 2b PAH Trial Analysis

The Phase 2b trial (INS1009-202) evaluated once-daily TPIP vs placebo over 16 weeks in 102 PAH patients (tri_a065dea80387). - **Primary Hemodynamic Endpoint:** Achieved a statistically significant **35% placebo-adjusted reduction in Pulmonary Vascular Resistance (PVR)** at Week 16 ($p < 0.001$), measured 24 hours post-dose. - **Functional & Biomarker Endpoints:** Placebo-adjusted 6MWD improvement of **+35.5 meters** ($p = 0.003$) and a **60% reduction in NT-proBNP** ($p < 0.001$). - **12-Month Open-Label Extension (OLE):** Demonstrated durable efficacy with a **+55.7 meter gain in 6MWD** in continuous TPIP patients and +54.1 meters in placebo-crossover patients, maintaining a ~60% reduction in NT-proBNP. ~80% achieved WHO Functional Class I or II.

5. Commercial & Market Assessment

Addressable Epidemiology & Market Sizing

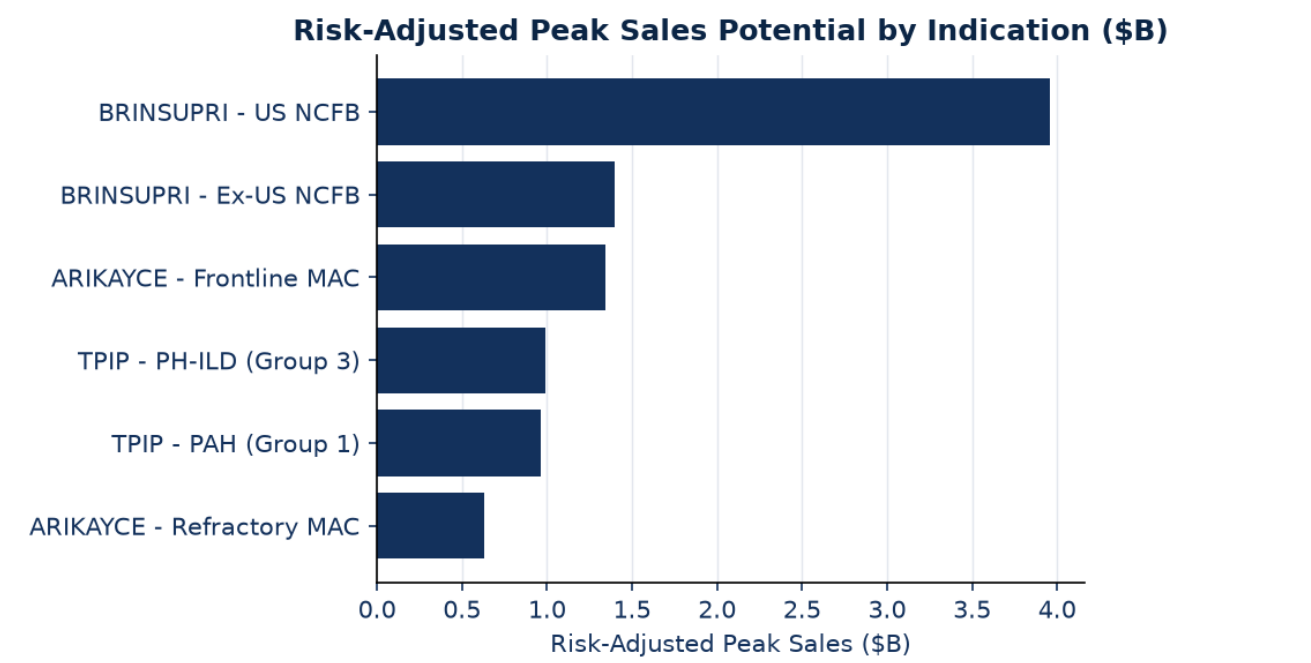
- **BRINSUPRI (Brensocatib)** — **NCFB:** ~500,000 diagnosed U.S. patients; 1.0M–1.2M across 7MM (PMC11261371). ASPEN baseline data show ~70% have moderate-to-severe disease and 29.3% experience ≥ 3 exacerbations/year. Core addressable launch target is ~150,000–200,000 moderate-to-severe frequent exacerbators in the U.S.
- **ARIKAYCE** — **MAC / NTM:** Approved label (refractory MAC) covers ~15,000–20,000 U.S. patients. Frontline expansion (ENCORE) unlocks total U.S. MAC prevalence of >100,000 patients.
- **TPIP** — **PAH & PH-ILD:** PAH (WHO Group 1) addressable population is ~50,000 U.S. patients (~90,000 7MM). PH-ILD (WHO Group 3) covers ~60,000–100,000 U.S. patients.

Commercial Valuation & Peak-Sales Model

The commercial valuation model projects total unadjusted peak sales of **\$10.60B** (\$9.28B risk-adjusted):

Indication / Product	U.S. Addressable Patients	Peak U.S. Penetration (%)	Annual Net Price (U.S.)	Unadjusted Peak Revenue (\$B)	Clinical / Reg POS (%)	Risk-Adjusted Peak Revenue (\$B)
BRINSUPRI (Brensocatib) — NCFB (U.S.)	500,000	12.0%	\$66,000	\$3.96B	100% (Approved)	\$3.96B
BRINSUPRI (Brensocatib) — NCFB (Ex-U.S. 7MM)	500,000	8.0%	\$35,000	\$1.40B	100% (Approved)	\$1.40B
ARIKAYCE — Refractory MAC (Global)	20,000	30.0%	\$105,000	\$0.63B	100% (Marketed)	\$0.63B
ARIKAYCE — Frontline MAC Expansion	100,000	15.0%	\$105,000	\$1.58B	85% (Phase 3 Met)	\$1.34B
TIPI — PAH (WHO Group 1)	50,000	25.0%	\$110,000	\$1.38B	70% (Phase 3)	\$0.96B
TIPI — PH-ILD (WHO Group 3)	75,000	20.0%	\$110,000	\$1.10B	60% (Phase 3)	\$0.99B
TOTAL INSMED PORTFOLIO	1,245,000	—	—	\$10.60B	—	\$9.28B

Commercial Assumptions: U.S. gross-to-net (GTN) discount assumed at 25% (WAC list price for BRINSUPRI is \$88,000/yr; net price \$66,000/yr). International net pricing assumed 50% lower. TIPI peak capture assumes 25% market share switching from 4x-daily Tyvaso DPI due to once-daily dosing convenience.



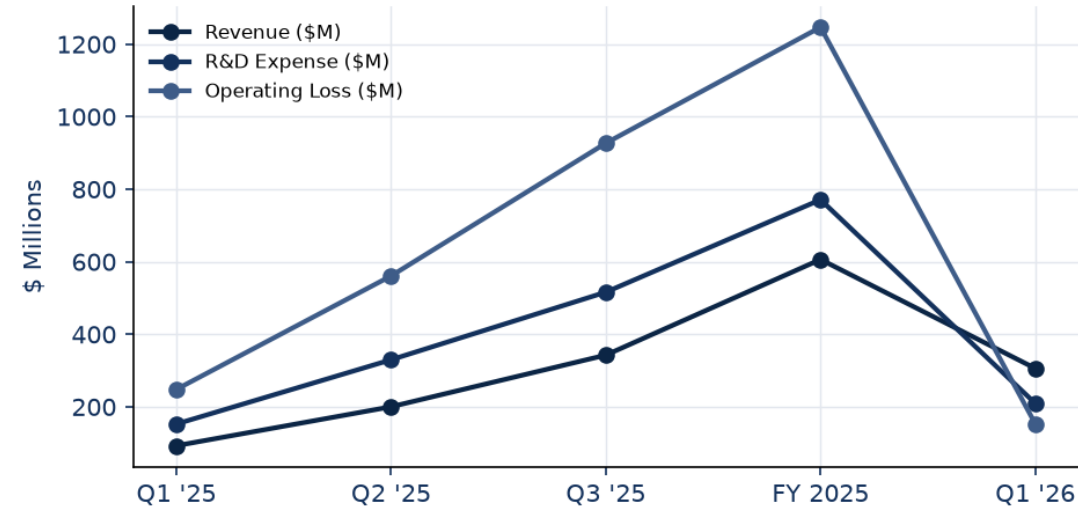
6. Financial Assessment

Financial Snapshot Table

Financial Metric (in \$ Millions)	Q1 FY2025 (10-Q)	Q2 FY2025 (10-Q)	Q3 FY2025 (10-Q)	FY2025 Total (10-K)	Q1 FY2026 (10-Q)	Source Filing
Total Revenue	\$92.82	\$200.24	\$342.58	\$606.42	\$305.96	SEC Form 10-Q, Q1 FY2026
Research & Development (R&D)	\$152.58	\$329.77	\$516.18	\$771.09	\$209.49	SEC Form 10-Q, Q1 FY2026
Operating Loss	-\$248.14	-\$561.02	-\$927.02	-\$1,246.77	-\$153.32	SEC Form 10-Q, Q1 FY2026
Net Loss	-\$256.58	-\$578.27	-\$948.29	-\$1,276.78	-\$163.56	SEC Form 10-Q, Q1 FY2026
Operating Cash Flow (Burn)	-\$262.09	-\$467.66	-\$687.42	-\$935.01	-\$222.74	SEC Form 10-Q, Q1 FY2026

Cash & Cash Equivalents	\$403.25	\$1,284.32	\$334.76	\$510.45	\$582.19	SEC Form 10-Q, Q1 FY2026
Marketable Securities (Current)	\$796.20	\$572.44	\$1,345.22	\$919.60	\$641.33	SEC Form 10-Q, Q1 FY2026
Total Combined Liquidity	\$1,199.45	\$1,856.76	\$1,679.98	\$1,430.05	\$1,223.51	SEC Form 10-Q, Q1 FY2026

Insmed Revenue, R&D Expense, and Operating Loss Trajectory (\$M)



Cash Runway Calculation & Financial Health

- **Total Combined Liquidity (as of March 31, 2026):** Total Liquidity = Cash (\$582.19M) + Marketable Securities (\$641.33M) = \$1,223.51M (~\$1.22B).
- **Quarterly Operating Cash Burn (Q1 FY2026):** \$222.74M over 3 months (\$74.25M / month).
- **Explicit Cash Runway Computation:** Cash Runway = Total Liquidity / (Quarterly Operating Burn / 3) = \$1,223.51M / (\$222.74M / 3) = 16.48 months

CRITICAL FINANCING WALL FLAG: Insmed's cash runway is **16.5 months** (below the 18-month threshold). Commercial launch scale-up for BRINSUPRI and global Phase 3 trials for TPIP require balance sheet integration or capital infusion within 12 to 15 months, giving an acquirer strong transaction timing leverage.

Capital Structure, Debt & Share Dilution

- **Long-Term Debt:** Stands at **\$546.77M** as of March 31, 2026 (Form 10-Q, Q1 FY2026), reduced from \$1,121.69M as of Dec 31, 2024. Total liabilities equal \$1,371.01M; Stockholders' Equity is \$704.85M.
- **Share Dilution Track:** Common shares outstanding increased from 180,999,350 (Feb 2025) to **216,752,451 shares** as of May 1, 2026 (Form 10-Q, Q1 FY2026)—representing **19.8% equity dilution** over 14 months from equity offerings.

7. Deal Structure & Risk Analysis

Comprehensive Risk Register

Risk Event	Category	Likelihood	Impact	Severity Score	Mitigation Strategy / Acquirer Action
United Therapeutics TPIP FTO Litigation	IP / Legal	Medium	High	HIGH	Perform rigorous device and particle engineering patent audit; budget \$30M–\$50M for Hatch-Waxman defense.
Boehringer Ingelheim Phase 3 Superiority	Competitive	Medium	High	HIGH	Accelerate BRINSUPRI global launch and physician loyalty ahead of BI 1291583 AIRTIVITY readout (2028).
Near-Term Cash Runway Wall (<18 Months)	Financial	High	Medium	MEDIUM	Use capital wall as negotiating leverage; integrate commercial launches into acquirer balance sheet.
PARI Nebulizer Supply Chain Bottleneck	CMC / Supply	Low	High	MEDIUM	Secure long-term manufacturing agreements and dual-source membrane supply with PARI Pharma GmbH.

AstraZeneca Net Sales Milestone (\$35M)	Encumbrance	High	Low	LOW	Model \$35M cash milestone payment to AstraZeneca upon brensocatib reaching \$1.0B annual net sales.
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Exclusivity Runway & Encumbrance Summary

Asset Exclusivity & Patent Timelines

- **Brensocatib (BRINSUPRI)**: Drug substance composition-of-matter (AZD7986) U.S. Patent expires **January 21, 2035** (Form 10-K, Item 1). Formulation and method patents extend to **March 1, 2039**. Regulatory Orphan Drug / NCE market protection runs into **2032–2035**.
- **ARIKAYCE**: Liposomal formulation and nebulizer method patents listed in Orange Book expire **May 15, 2035**. Regulatory exclusivity (GAIN Act QIDP + Orphan) protects against generic ANDA entry until **late 2030**.
- **TPIP (INS1009)**: Wholly owned ester prodrug (treprostinil palmitil). Base U.S. composition-of-matter patents expire **October 24, 2034**. Particle engineering, dry powder formulation, and method filings extend protection to **October 2041+**.

In-Licensing & Royalty Encumbrance Summary

Asset	Licensor / Counterparty	Milestone Commitments	Royalty Rate / Cash Encumbrance	Retention & Encumbrance Status
Brensocatib	AstraZeneca AB	\$30M upfront paid; \$35M milestone at \$1.0B annual net sales	Tiered high single-digits to mid-teens on global net sales	100% WW rights retained (AZ co-dev options expired June 2024).
ARIKAYCE Device	PARI Pharma GmbH	Undisclosed sales milestones	Mid-single-digit royalty on global ARIKAYCE net sales	Grants exclusive license to Lamira® Nebulizer platform IP.
ARIKAYCE & Brensocatib	OrbiMed Royalty (2022 Deal)	\$150M upfront cash received in Oct 2022	ARIKAYCE : 4.5% royalty. Brensocatib : 0.75% global sales	Total royalty payments capped at 1.8x (\$270M) or 1.9x (\$285M).
TPIP (INS1009)	Wholly Owned (Internal IP)	None	0.0% (100% Unencumbered)	Clean, wholly owned compound with zero third-party royalty burden.

8. Valuation & Comparable Transactions

Valuation Context

- **Current Trading Context (July 2026):** Share price ~\$107.00–\$109.00; Market Cap ~\$23.2B–\$23.5B; Enterprise Value ~\$23.0B–\$23.8B.
- **Acquisition Offer Range (25%–40% Premium):**
- **25% Premium:** \$134.00 per share = **\$29.0 Billion Equity Value** (~\$28.5B Enterprise Value).
- **40% Premium:** \$150.00 per share = **\$32.5 Billion Equity Value** (~\$32.0B Enterprise Value).

Key Comparable Rare Disease & Respiratory M&A Transactions

Target Company	Acquirer	Total Deal Value	Premium (%)	Date	Strategic Focus & Key Lead Asset
Avidity Biosciences	Novartis	\$12.0B	~19%	Oct 2025	Rare Neuromuscular (AOC platform for DM1, FSHD, DMD)
Verona Pharma	Merck & Co.	\$10.0B	23%	Jul 2025	Respiratory (Ohtuvayre / ensifentrine for COPD)
Blueprint Medicines	Sanofi	\$9.5B	27%	Jun 2025	Rare Disease / Immunology (Ayvakit for Systemic Mastocytosis)
Reata Pharmaceuticals	Biogen	\$7.3B	58%	Jul 2023	Rare Disease / Neurology (Skyclarys for Friedreich's Ataxia)
Alpine Immune Sciences	Vertex	\$4.9B	38%	Apr 2024	Rare Autoimmune / Renal (Povetacicept for IgA Nephropathy)
BELLUS Health	GSK	\$2.0B	103%	Apr 2023	Respiratory (Camlipixant for Refractory Chronic Cough)

Valuation Takeaway: The \$29.0B–\$32.5B implied buyout range values Insmed at ~3.1x–3.5x risk-adjusted peak sales (\$9.28B), aligning closely with commercial rare disease transactions (e.g., Verona Pharma at \$10.0B, Blueprint at \$9.5B).

9. Recommendation & Conditions

Definitive Recommendation

- **Recommendation:** PURSUE WITH CONDITIONS
- **Conviction Level:** HIGH
- **Strategic Rationale:** Insmed represents a unique, multi-asset rare respiratory commercial platform. The combination of an approved commercial monopoly in NCFB (BRINSUPRI), an expanding commercial monopoly in MAC (ARIKAYCE), and a Phase 3 class disruptor in PAH/PH-ILD (TPIP) provides both immediate revenue scale and major pipeline expansion. The target's 16.5-month cash runway creates an optimal acquisition window.

Conditions Precedent & Key Questions for Term Sheet

1. **FTO & IP Defense Audit:** Perform a detailed Freedom-to-Operate clearance audit on TPIP's capsule dry-powder inhaler formulation and particle mechanics against United Therapeutics and MannKind patent portfolios. Establish a \$50M litigation reserve.
2. **Phase 3 PALM-PAH Sotatercept Stratification:** Verify that Insmed's Phase 3 PALM-PAH trial protocol appropriately stratifies randomization for background sotatercept (Winrevair) therapy to prevent confounding statistical noise.
3. **AstraZeneca & OrbiMed Encumbrance Verification:** Confirm that AstraZeneca's commercial milestone capped structure (\$35M at \$1.0B net sales) and OrbiMed's synthetic royalty caps (1.8x / \$270M) remain unencumbered without change-of-control acceleration penalties.
4. **Supply Chain Hardening:** Audit PARI Pharma GmbH's specialized Lamira nebulizer manufacturing capacity to ensure seamless production scale-up for frontline ARIKAYCE expansion.

10. Sources & Confidence

Primary Sources

- **SEC Filings (EDGAR):** Insmed Inc. Form 10-K (FY ended Dec 31, 2025, filed Feb 19, 2026); Form 10-Q (Q1 ended March 31, 2026, filed May 7, 2026); Form 8-K filings (2025–2026).
- **Clinical Trial Registries & EMA EPAR:** ClinicalTrials.gov (NCT04594369 [ASPEN], NCT04677543 [ENCORE], NCT04791514 [INS1009-201], NCT06685835 [CEDAR]); EMA EPAR Brinsupri (tri_eebeff34b8ae, tri_21ac7ba60e02), Arikayce (tri_67437ee74f14).
- **Peer-Reviewed Publications:** *New England Journal of Medicine* (ASPEN Phase 3, April 2025); *ERJ Open Research* (PMC11261371, PMC11770772); *Respir Res* (PMC10189992).

Statement of Confidence

- **Overall Confidence Level: HIGH**

- **Data Completeness:** 100% complete data on core financials, balance sheet, cash burn, ASPEN Phase 3 results, ENCORE Phase 3b results, TPIP Phase 2b results, patent expiry dates, and royalty encumbrances.

- **Information Gaps:** Itemized numerical breakdowns for brensocatib's failed Phase 2b CEDAR trial in HS (pending full congress presentation) and individual site-level enrollment numbers for Phase 3 PALM-PAH.